

Many Vortices from a Migrating Planet

A Rossby-Wave-Like Linear Instability

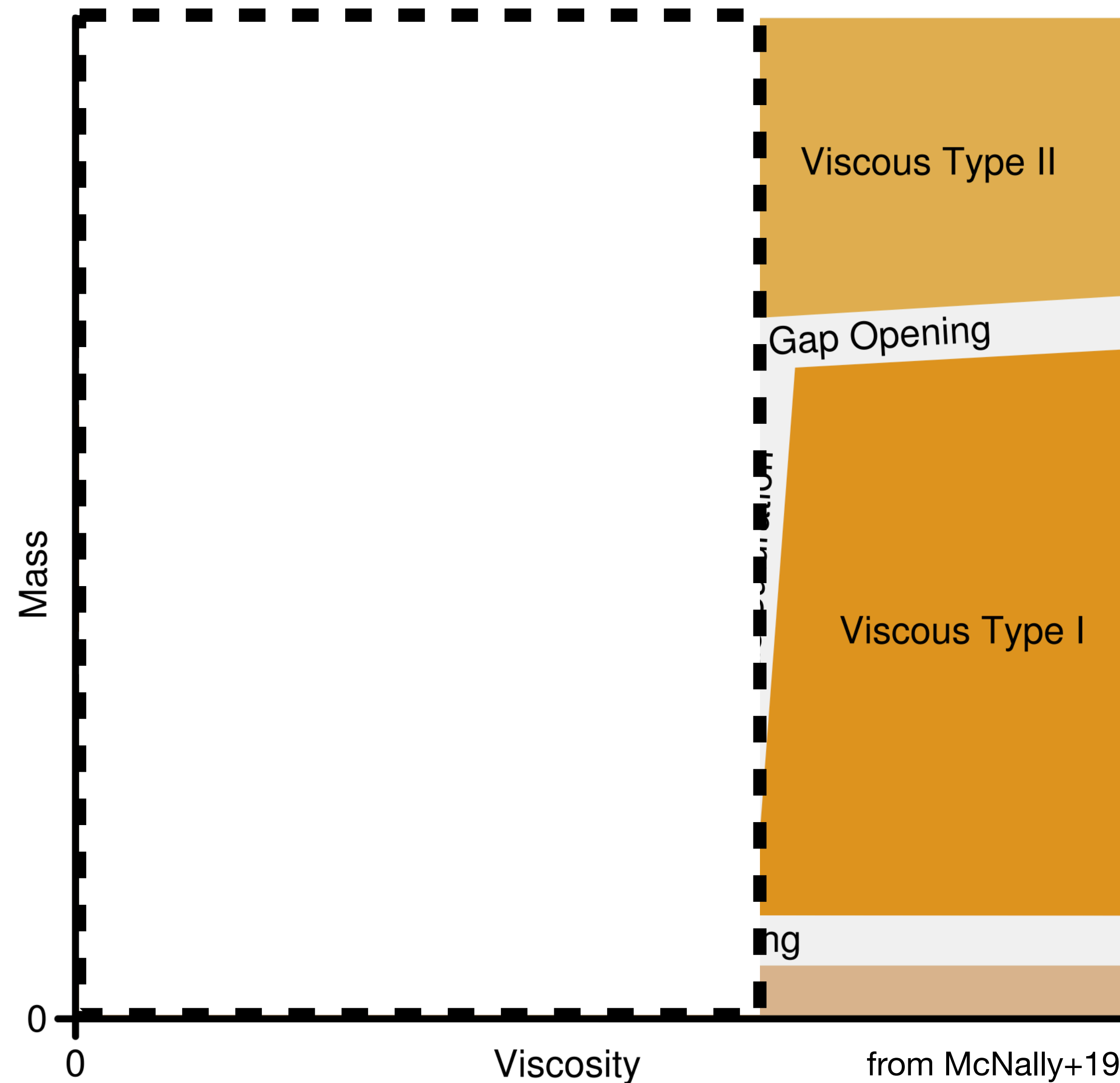
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FARGO3D simulation

Planet migration in low-viscosity disks

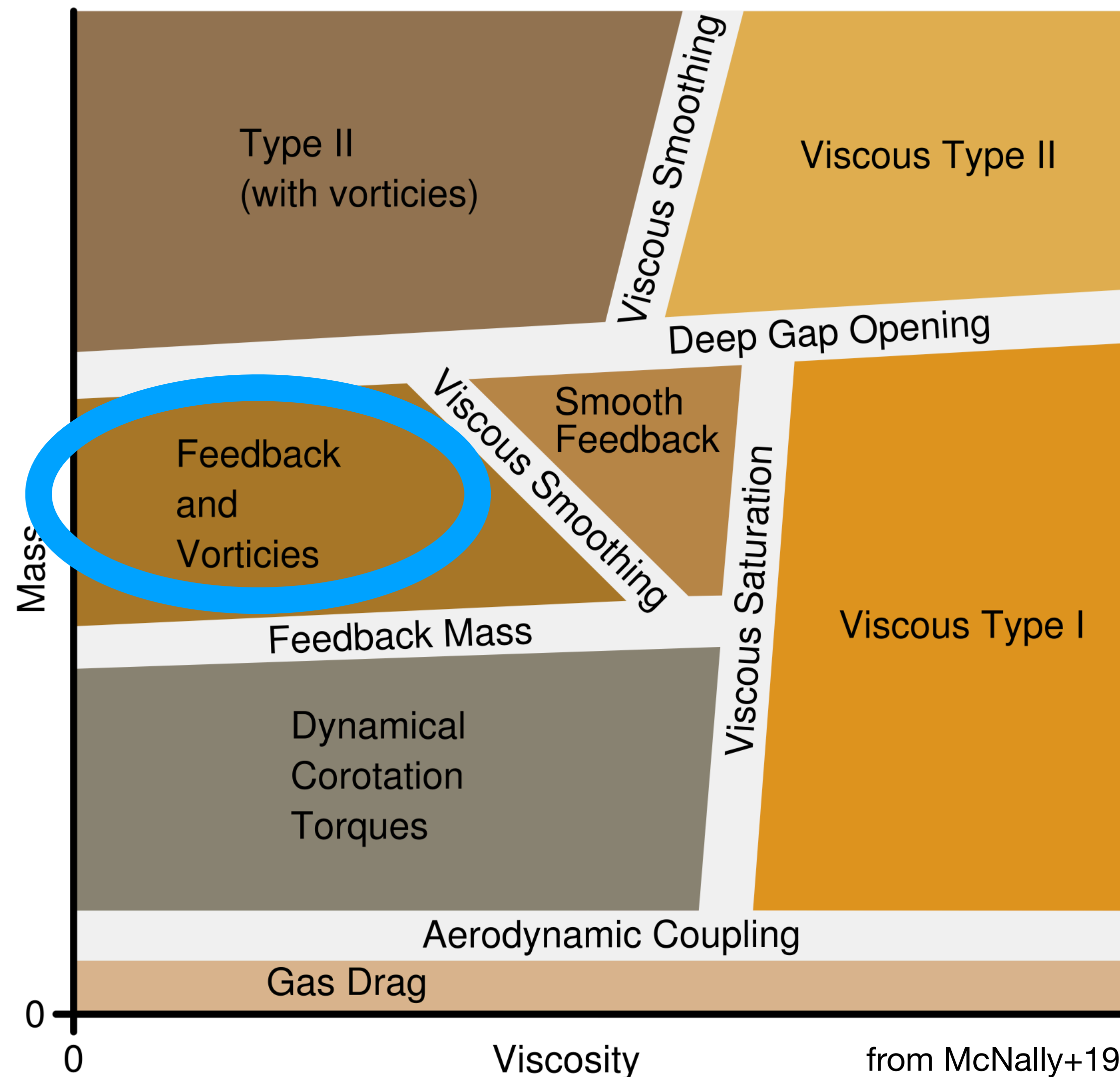


- Type II: deep gap and slow down
(e.g. Lin & Papaloizou 86)

- Type I: rapid inward migration
(e.g. Goldreich & Tremaine 79)

Planet migration in low-viscosity disks

- For $\alpha \lesssim 10^{-4}$, **vortices** modify planet migration (e.g. Lin & Papaloizou 10)
- can even **drive migration**

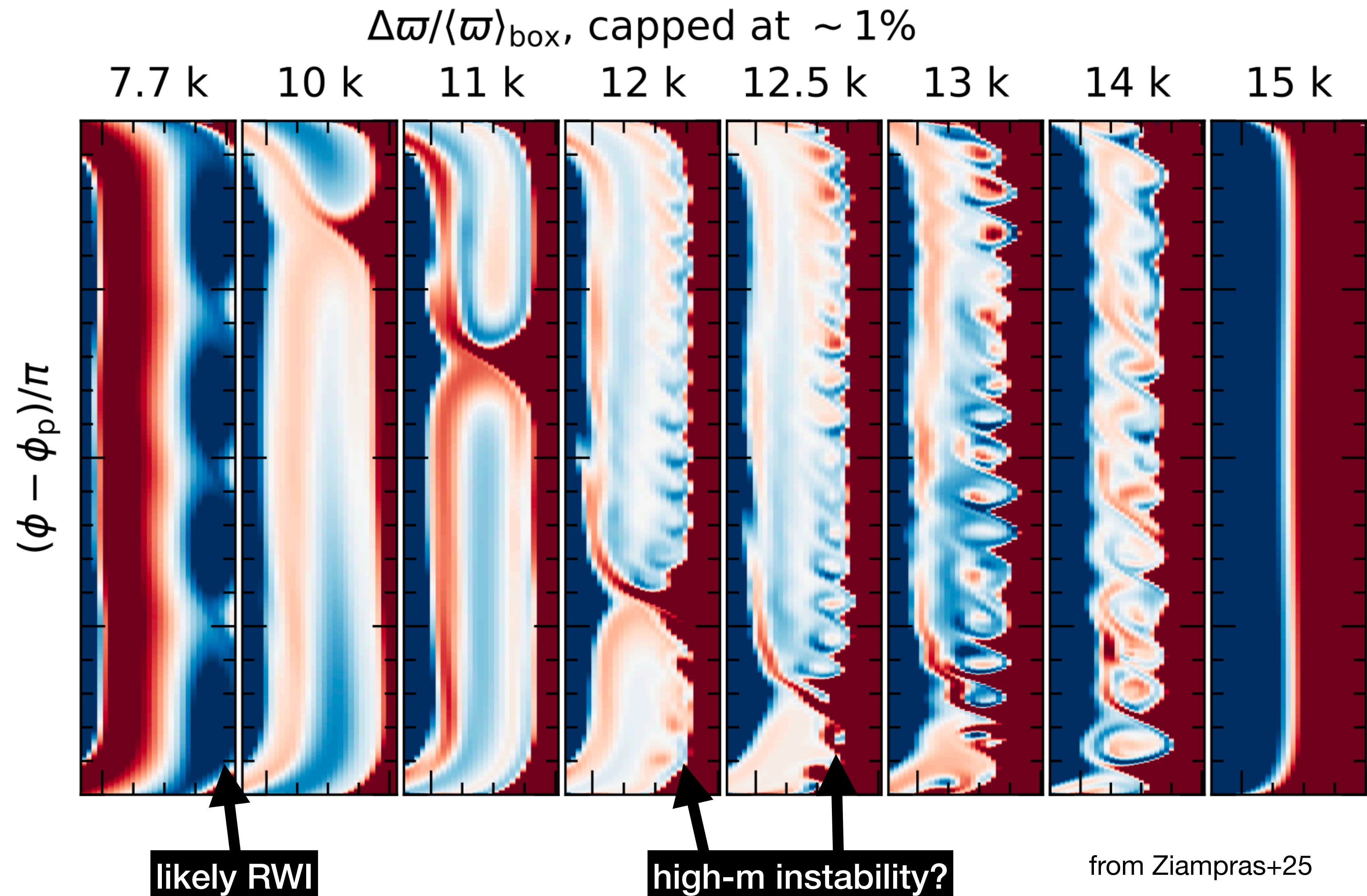


- Type II: deep gap and slow down (e.g. Lin & Papaloizou 86)

- Type I: rapid inward migration (e.g. Goldreich & Tremaine 79)

Why so many vortices?

- Recent simulations show 10-20 vortices (McNally+19, Ziampras+25)
- Classical RWI spawns 3-7 vortices
- **Is there another instability?**
- Try linear instability



Testing the linear instability hypothesis

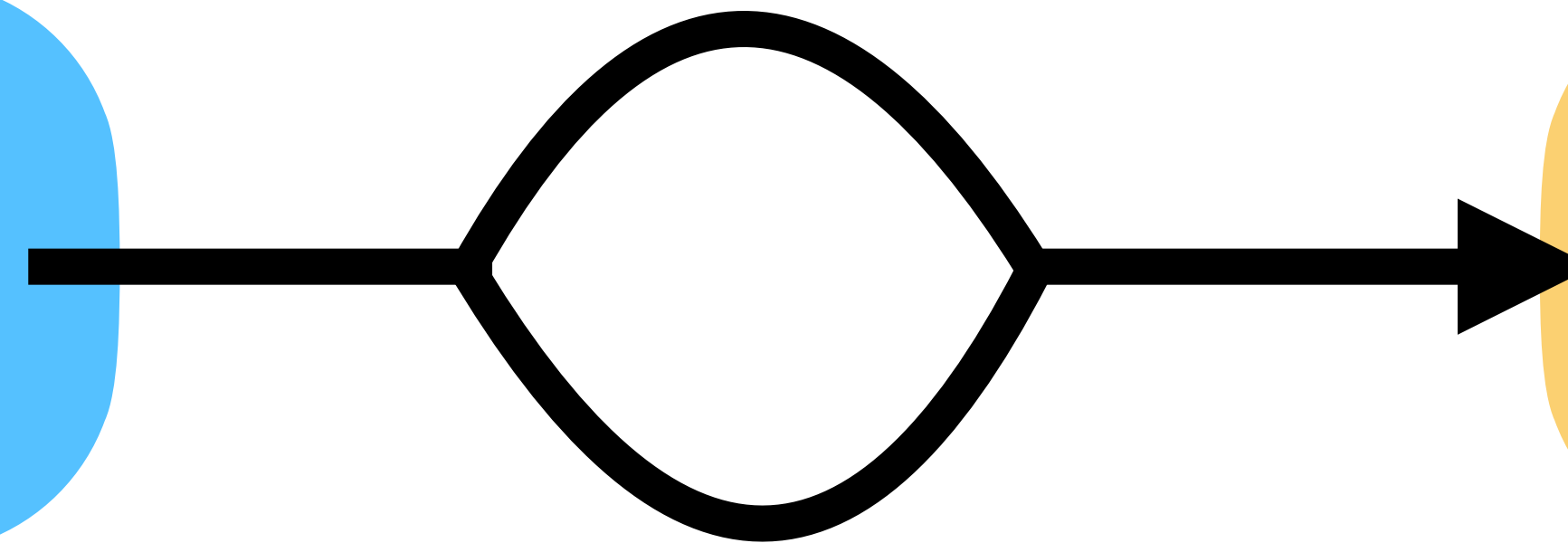
- Extract "equilibrium"

- Run simulation

Inviscid, globally isothermal

$$M_p = 0.5M_{th}$$

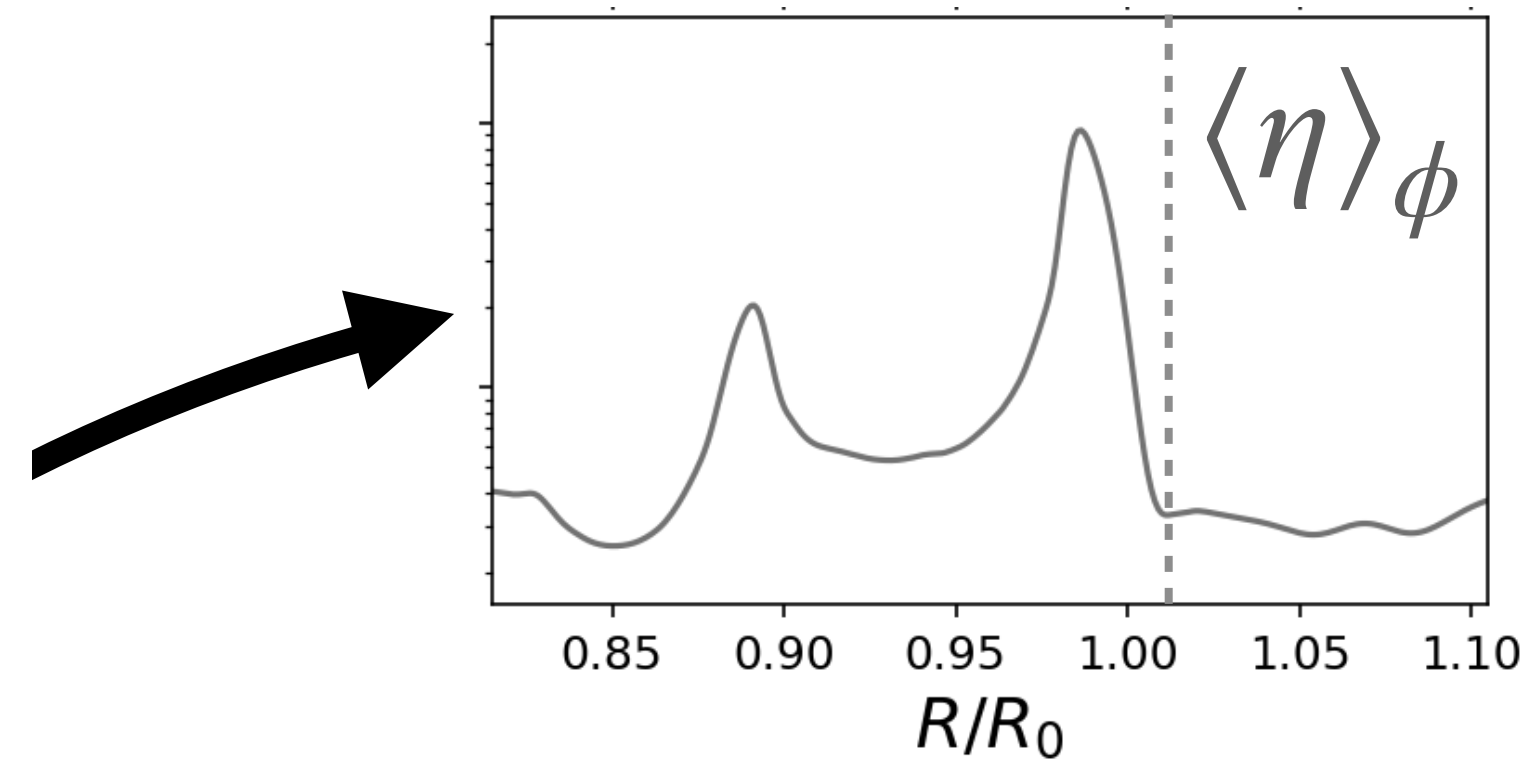
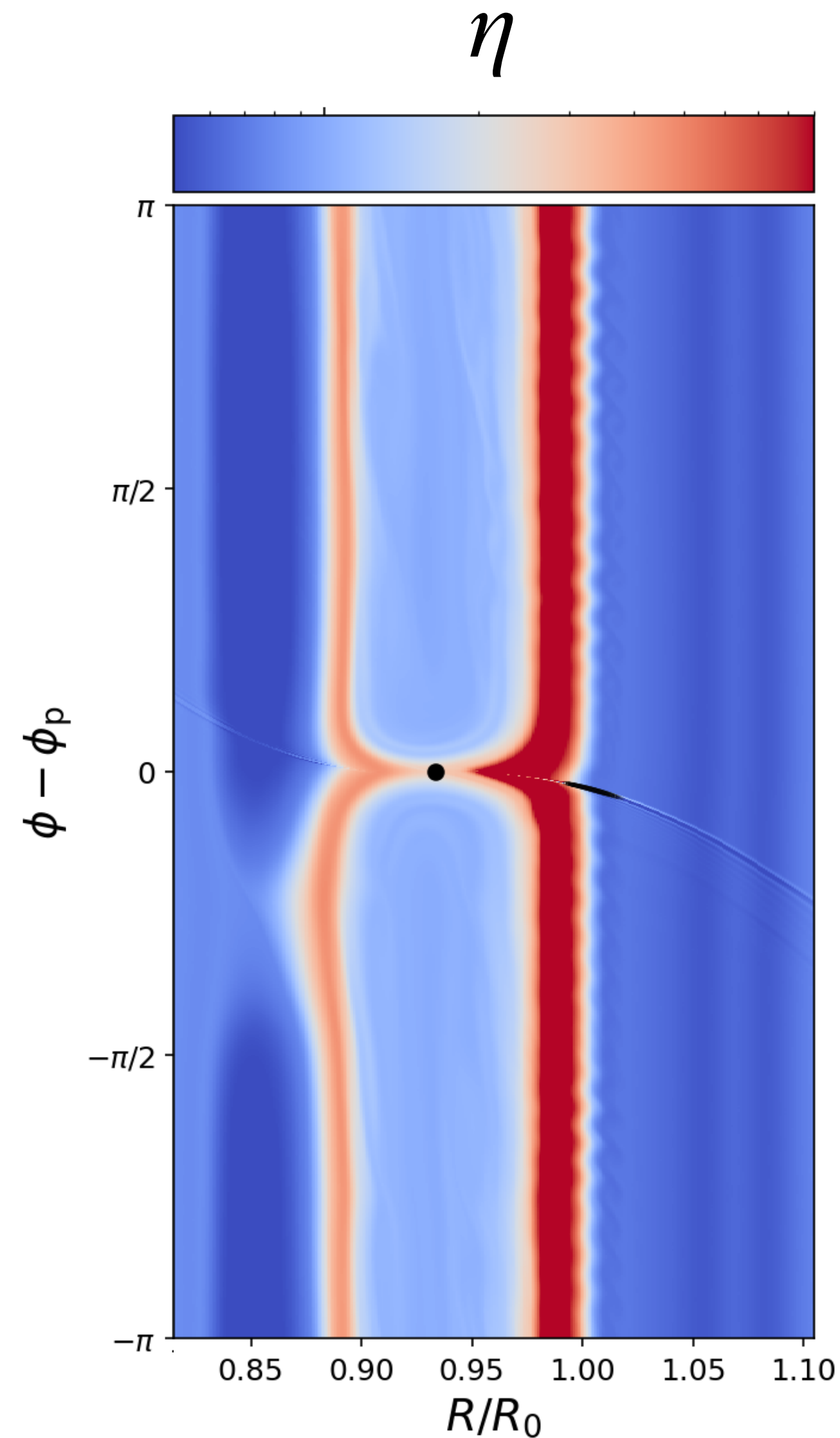
$$H/R = 0.025$$



- **Analyze stability
of perturbations**

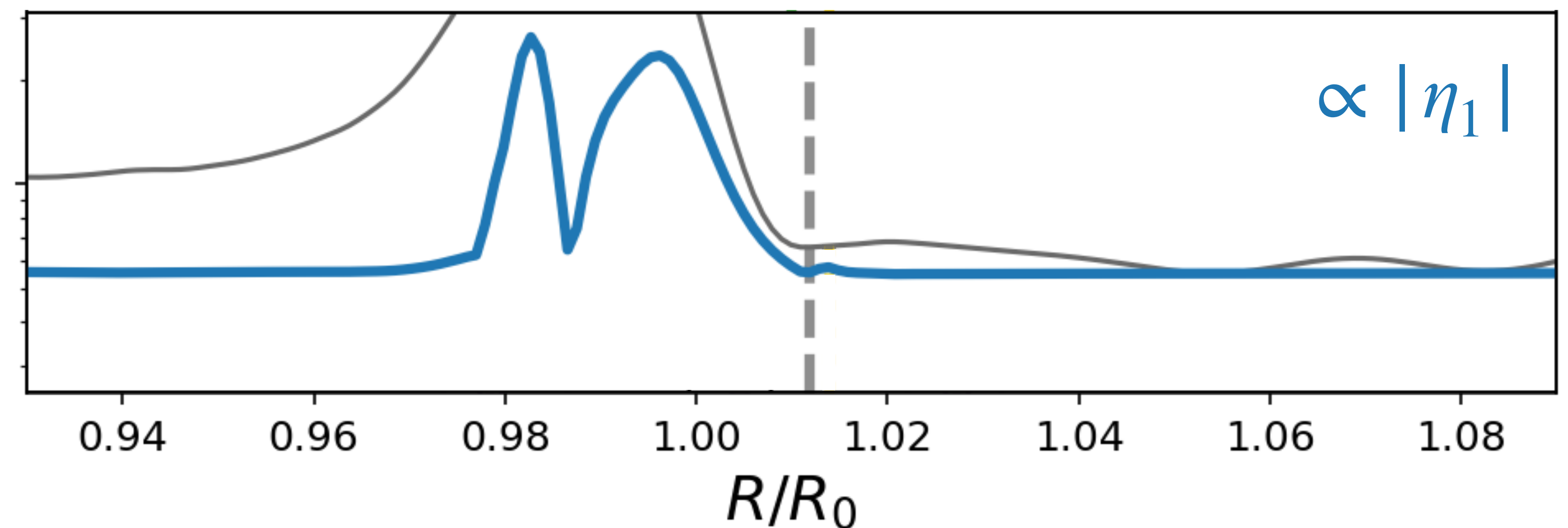
- Compute linear
perturbations

From simulation to linear analysis



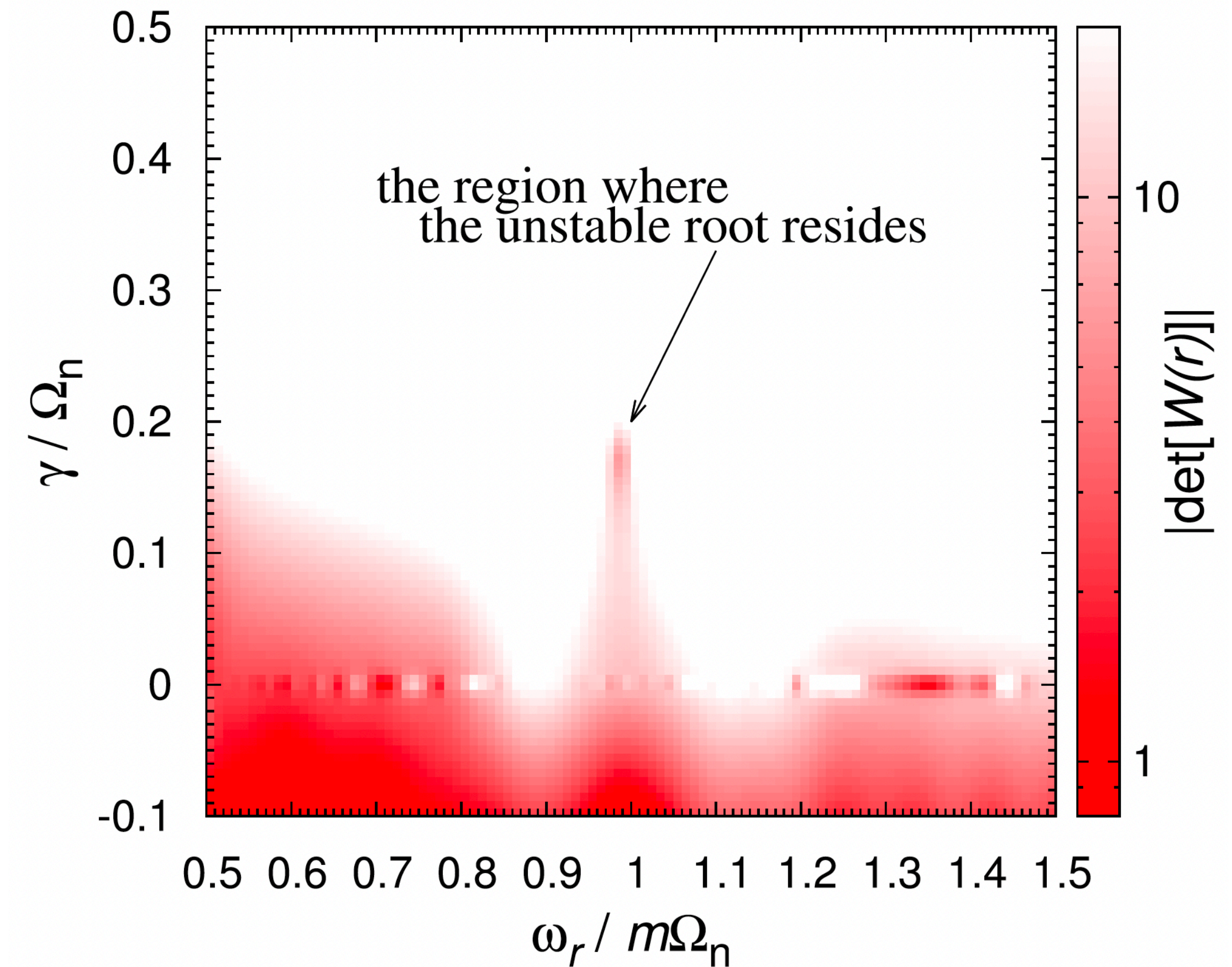
- Analyze each snapshot as an axisymmetric equilibrium

- $\eta \approx \langle \eta \rangle_\phi + \eta_1$



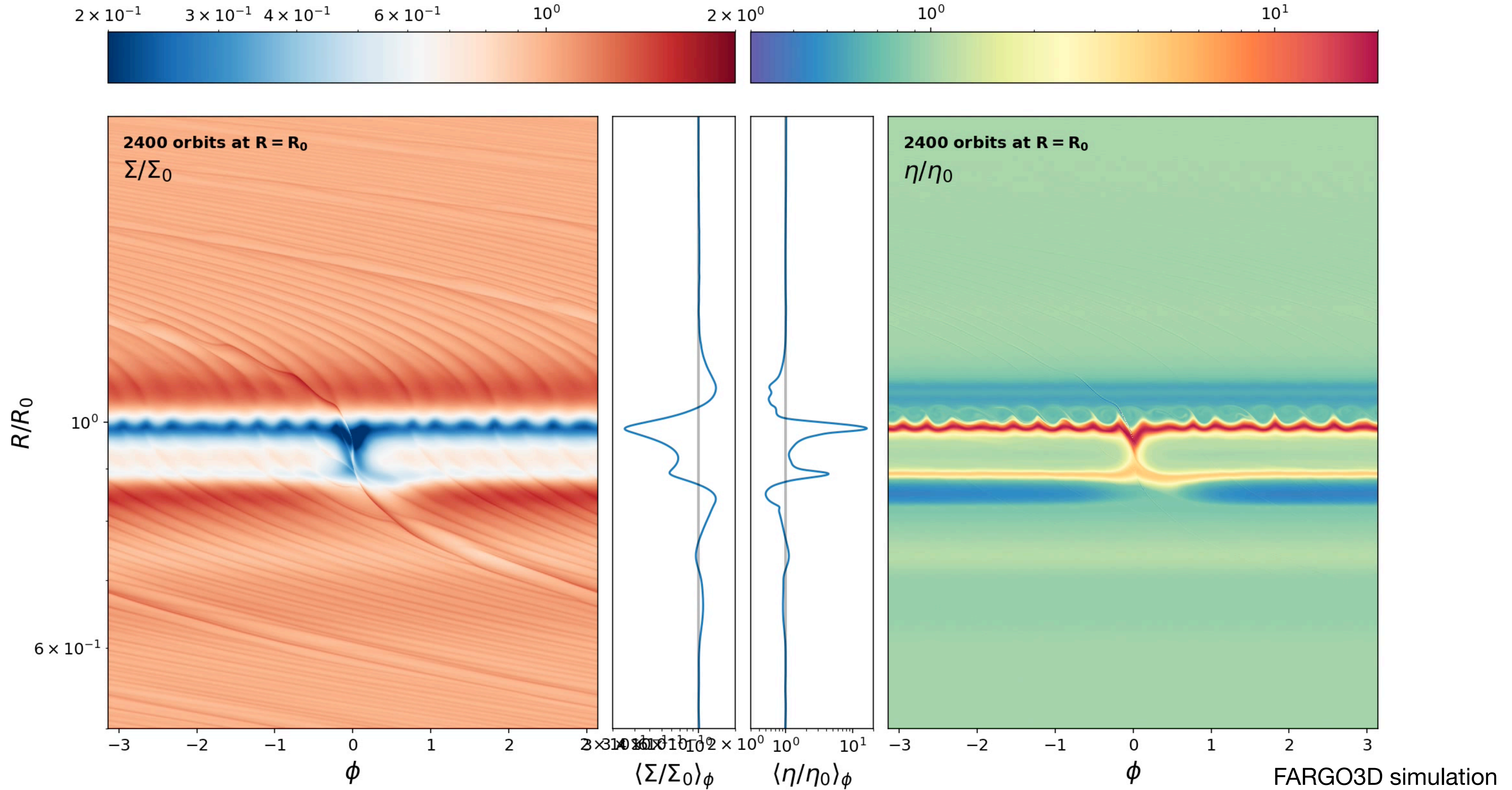
Linear stability analysis

- $\eta \approx \langle \eta \rangle_\phi + \eta_1, \quad \eta_1 \propto \exp[im\phi - i\omega t]$
 - m : azimuthal mode number
 - ω_r : wave frequency
 - ω_i : growth rate
 - $\omega_i > 0 \implies$ instability
- Find ω satisfying the linearized equations:
$$\Psi'' + B\Psi' + C\Psi = 0$$



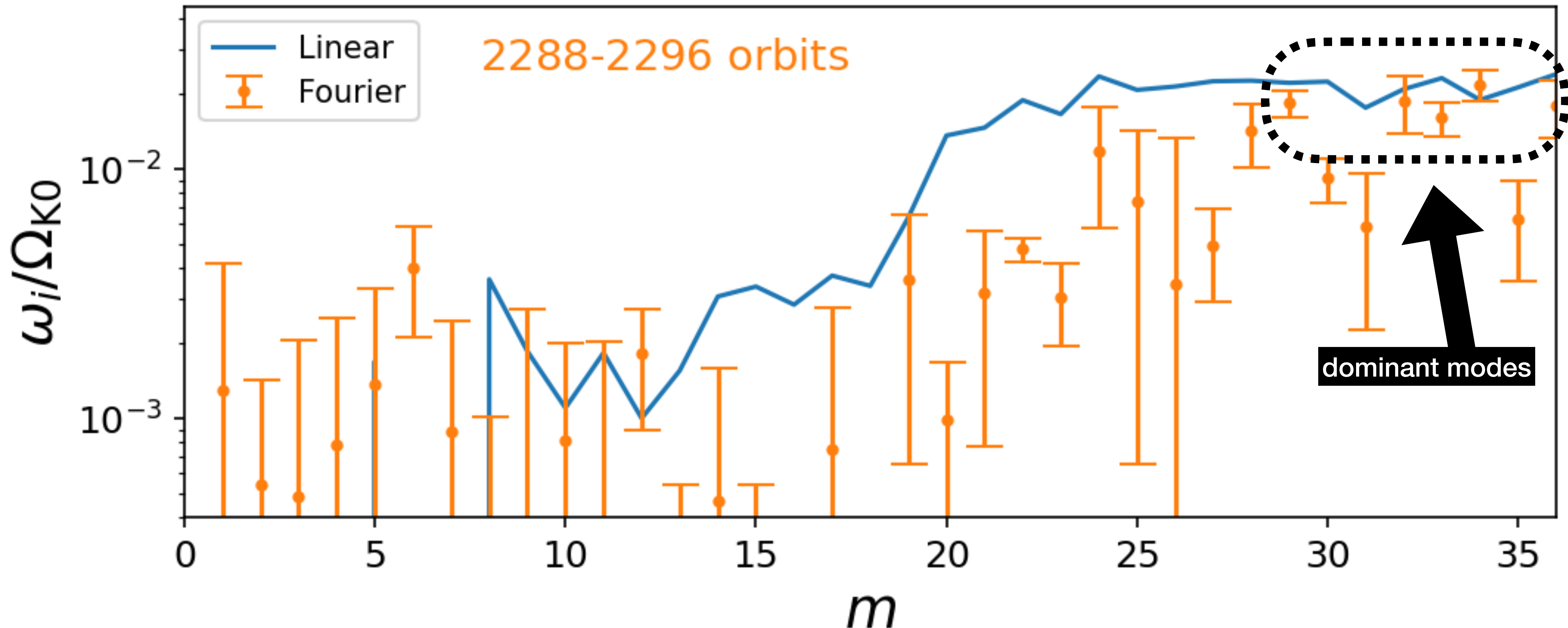
from Ono+16

Many vortices triggered by a migrating planet

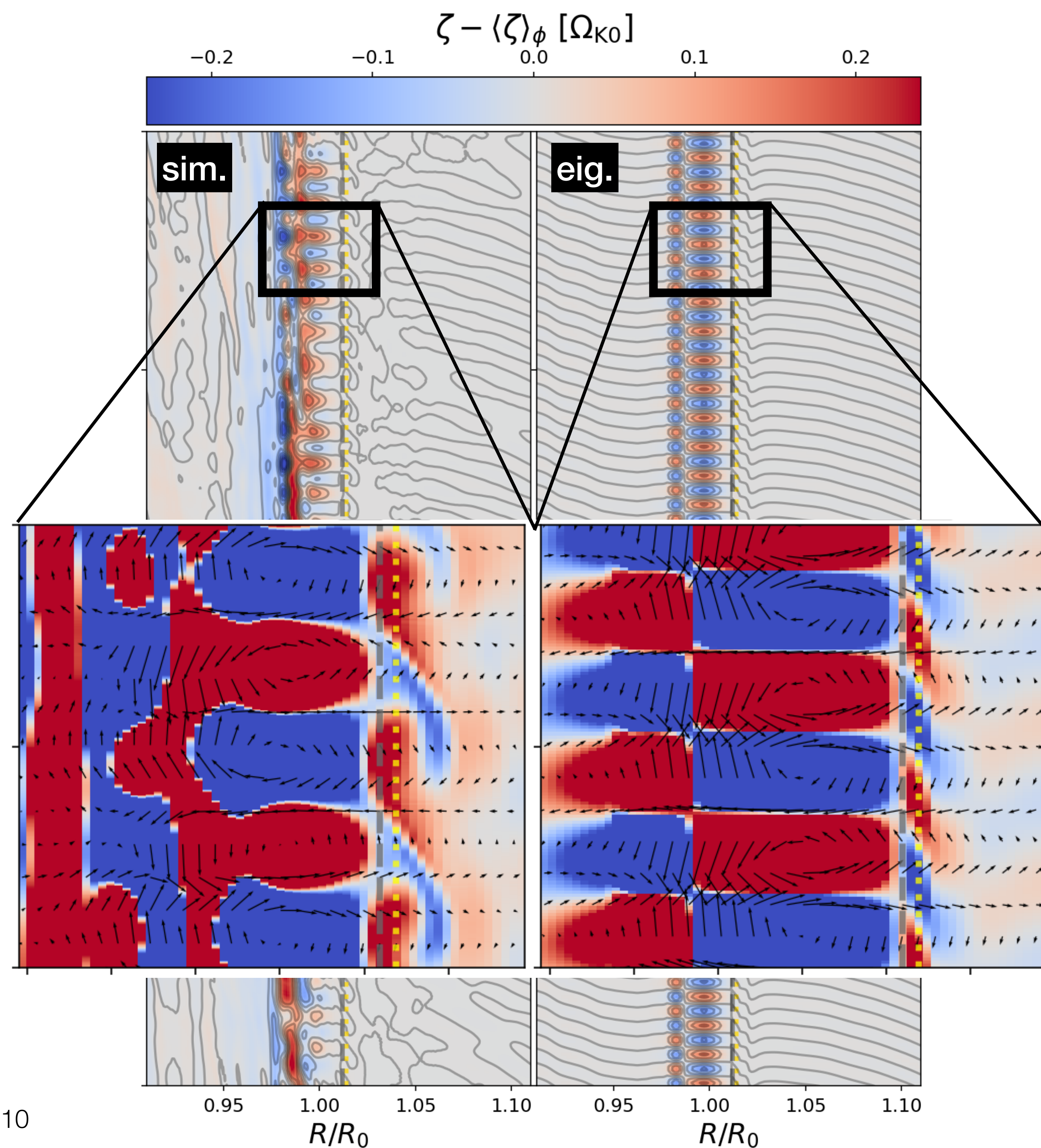


Observed growth agrees with linear theory

- Linear theory predicts dominant modes



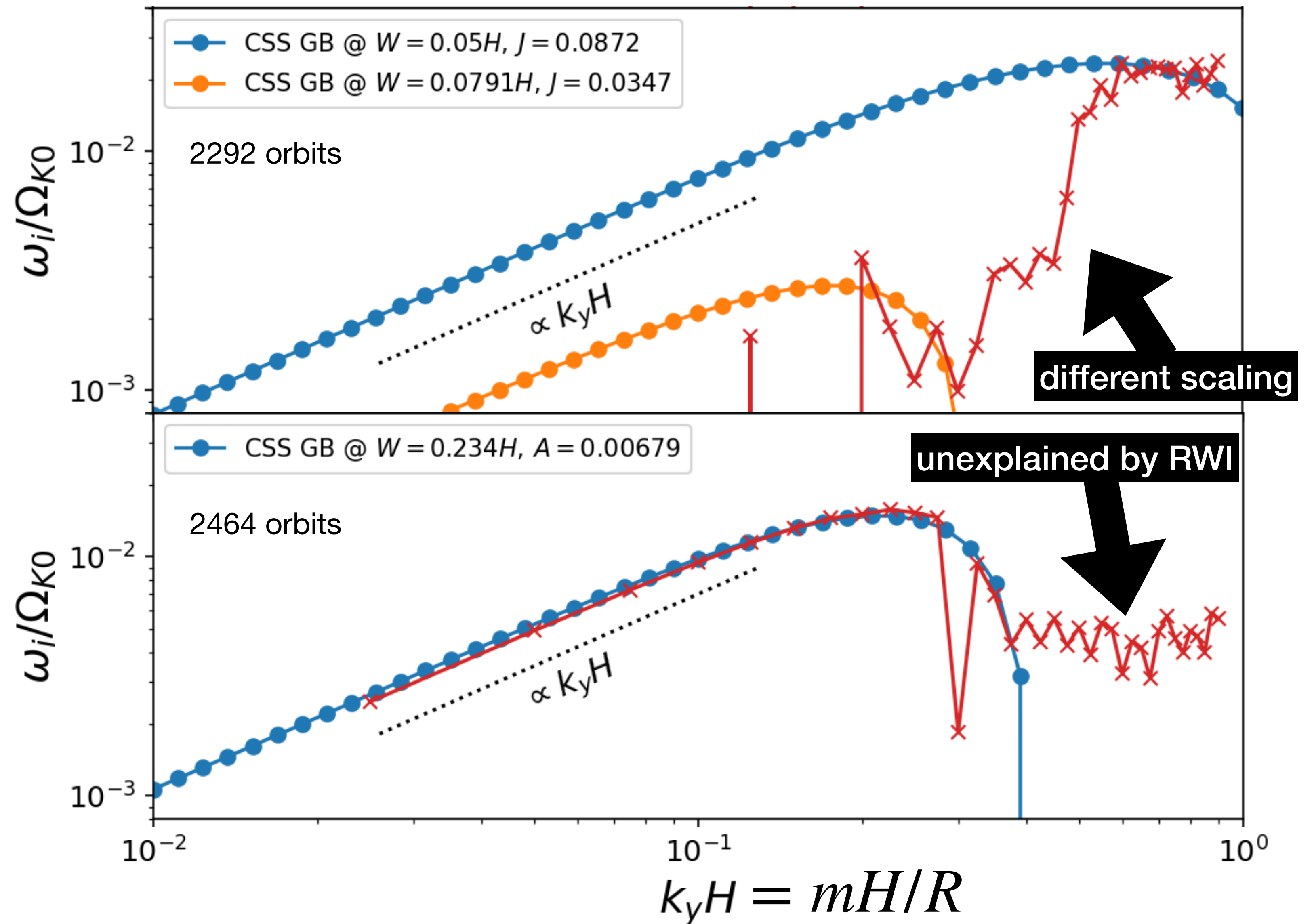
Linear modes match vortex morphology



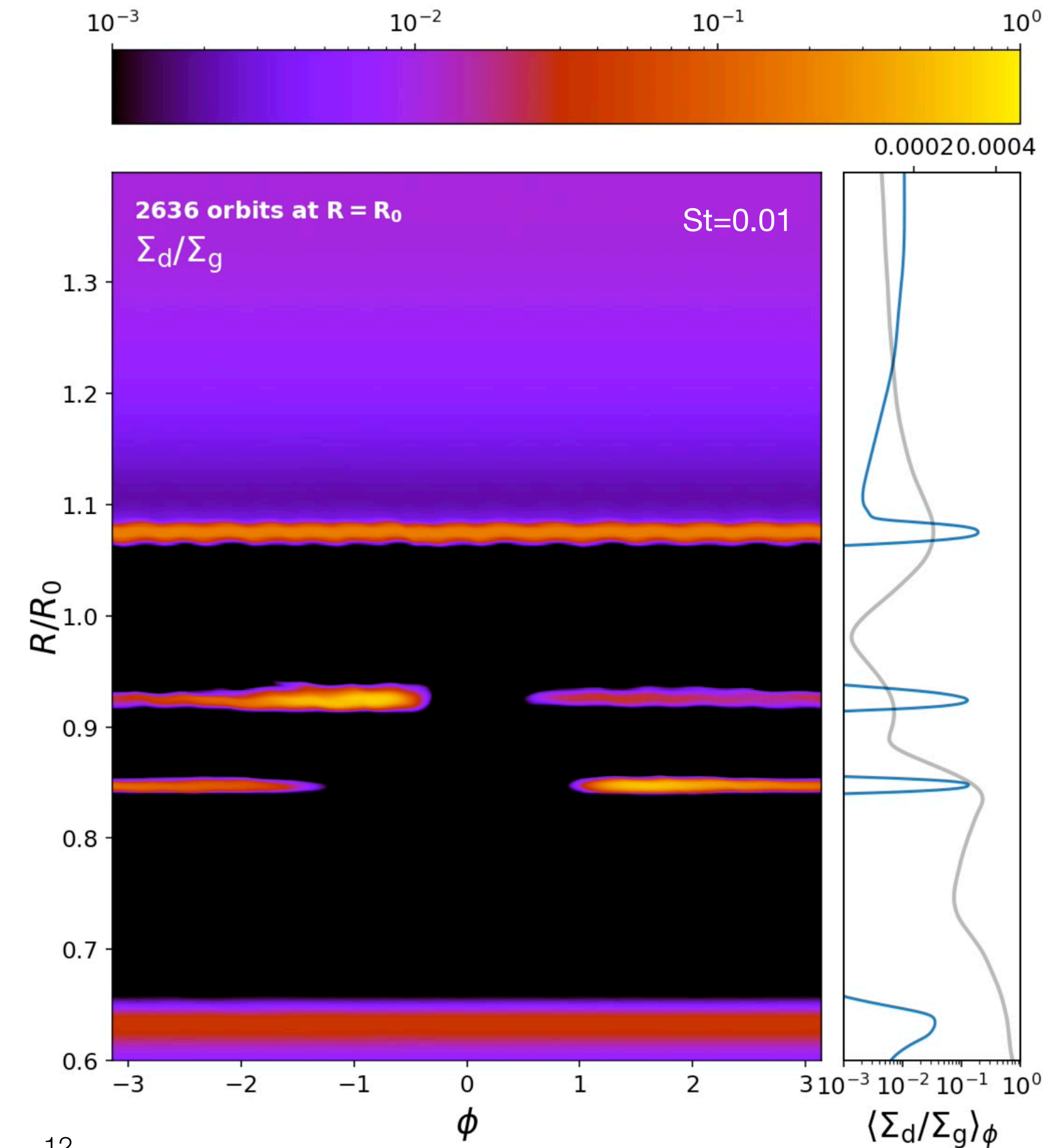
- **FGM: $m=33 \rightarrow \sim 30$ vortices**
- Linear mode reproduces simulated pattern
- Growth occurs via counter-propagating Rossby waves

RWI-like, but not classical RWI

- High- m instability does **not** follow the classical RWI dispersion relation
- growth does not scale as $k_y H$
- growth persists beyond RWI cutoff



...instability may be hard to see



- Dust is trapped in a ring to the exterior of vortices
- No dust reaches vortices
- Instability produces weak corrugations in the exterior ring
- **Likely difficult to resolve spatially**

Conclusion

- **A Rossby-like instability** can form ~30 vortices in low-viscosity disks ($\alpha \lesssim 10^{-5}$)
- Linear growth rates match Fourier growth rates
- Like RWI, growth occurs via counter-propagating Rossby waves
- Dispersion relation is different from that of classical RWI
- Dust does not trace the vortices, making it difficult to observe

